

Introduction to Probabilistic Machine Learning

Ralf Herbrich, Rainer Schlosser

Inference & Decision Making

Overview

1. Inference Methods
 - Bayesian Inference
 - Maximum Likelihood Estimation
2. Decision Making

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1. Inference Methods

- Bayesian Inference
- Maximum Likelihood Estimation

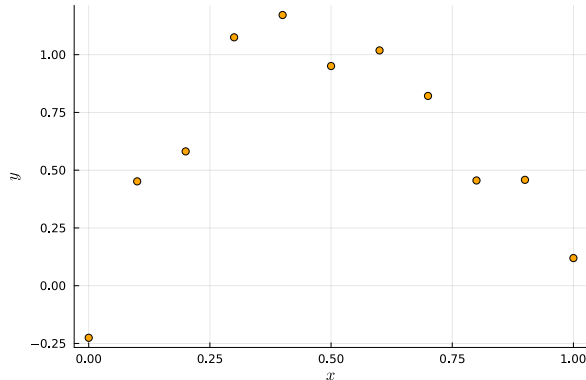
2. Decision Making

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Probabilistic Machine Learning: Ingredients

- 1. Training Data:** $D \in (\mathcal{X} \times \mathcal{Y})^n$
of n (labelled) examples from the input space $\mathcal{X}$ and output space $\mathcal{Y}$

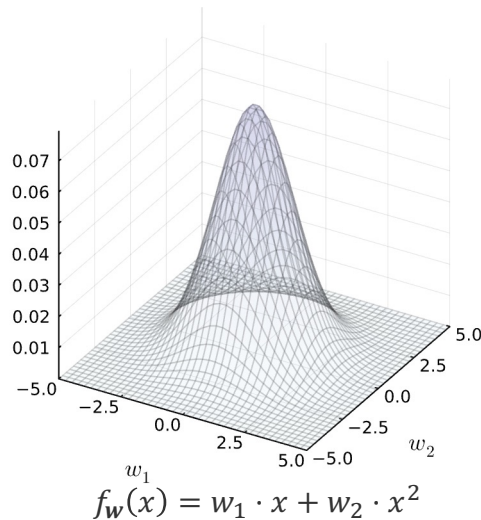


Training Data

$$D \subset \mathbb{R}^2$$

- 2. Prior belief over functions from $\mathcal{X}$ to $\mathcal{Y}$:** $p(f)$, $f \in \mathcal{F}$

- Space of functions, $\mathcal{F}$, is also called *hypothesis space*.

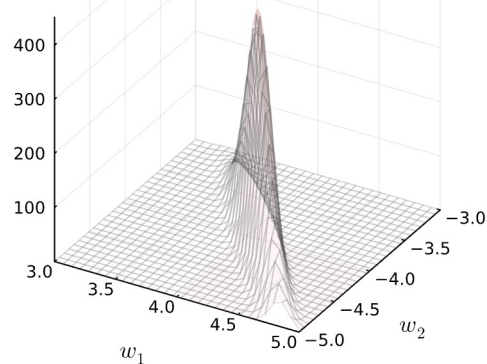


Prior

$$p(f_w) = \mathcal{N}(w_1; 0, 1) \cdot \mathcal{N}(w_2; 0, 1)$$

- 3. Likelihood of function:** $p(D|f) =: \ell(f)$

- Link between data and functions
- Models all assumptions how data/labels are generated from a function



Likelihood

$$\ell(f_w) = \prod_i \mathcal{N}(y_i; w_1 x_i + w_2 x_i^2, \sigma^2)$$

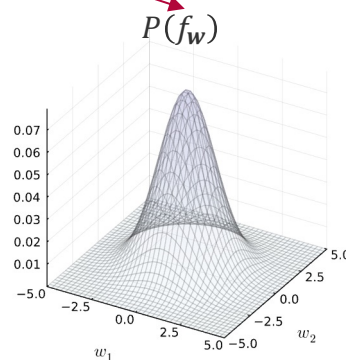
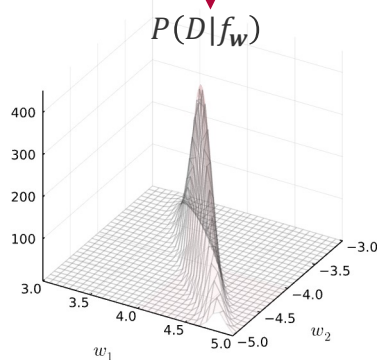
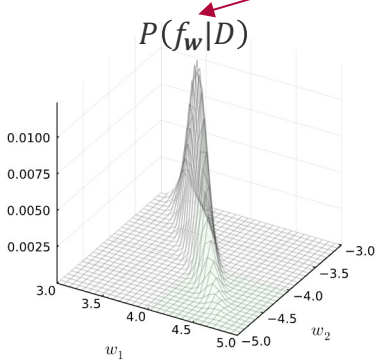
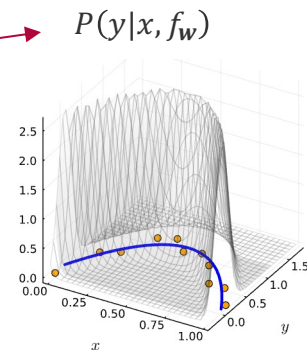
Key Question I: Inference and Predictive Distribution

- **Predictive Distribution.** Given a training set $D \in (\mathcal{X} \times \mathcal{Y})^n$ and a new input point $x \in \mathcal{X}$, the distribution $p(y|x, D) = \int p(y|x, f) \cdot p(f|D) df$ of target values $y \in \mathcal{Y}$ at the input point x is called the predictive distribution.

Total probability rule

- **Observation 1:** For any function $f \in \mathcal{F}$ from the hypothesis space, the likelihood is already the distribution $p(y|x, f)$!
- **Observation 2:** Each function $f \in \mathcal{F}$ from the hypothesis space has a posterior belief $p(f|D)$ after we have observed the training data using Bayes' rule!

$$p(f|D) = \frac{p(D|f) \cdot p(f)}{p(D)}$$



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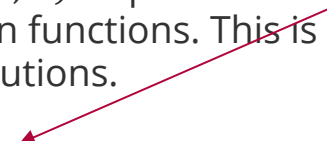
Probabilistic Machine Learning: Bayesian Inference

■ Two computational difficulties:

1. **Posterior** $p(f|D)$ requires the *multiplication* of likelihood with prior which often results in a distribution which is no longer in a family with very few parameters.

$$p(f|D) = \frac{p(D|f) \cdot p(f)}{p(D)} \propto \ell(f) \cdot p(f)$$


2. **Predictive distribution** $p(y|x, D)$ requires the *summation* of the data distribution over all prediction functions. This is only feasible for a small number of parametric distributions.

$$p(y|x, D) = \int p(y|x, f) \cdot p(f|D) df$$


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Probability Distributions: Conjugacy

- **Bayes Rule for Random Variables.** For any probability distribution p over two random variables X and Θ , it holds

$$\text{Posterior} \rightarrow p(\theta|x) = \frac{\text{Likelihood } p(x|\theta) \cdot \text{Prior } p(\theta)}{p(x)}$$

- **Conjugacy.** A family $\{p(x, \theta)\}_{x, \theta}$ is conjugate if the posterior $p(\theta|x)$ is part of the same family as the prior $p(\theta)$ for any value of x .

Likelihood $p(x \theta)$	Prior $p(\theta)$	Posterior $p(\theta x)$
$\text{Ber}(x; \theta)$	$\text{Beta}(\theta; \alpha, \beta)$	$\text{Beta}(\theta; \alpha + x, \beta + (1 - x))$
$\text{Bin}(x; n, \theta)$	$\text{Beta}(\theta; \alpha, \beta)$	$\text{Beta}(\theta; \alpha + x, \beta + (n - x))$
$\mathcal{N}(x; \theta, \sigma^2)$	$\mathcal{N}(\theta; m, s^2)$	$\mathcal{N}\left(\theta; x \cdot \frac{s^2}{s^2 + \sigma^2} + m \cdot \frac{\sigma^2}{s^2 + \sigma^2}, s^2 \cdot \frac{\sigma^2}{s^2 + \sigma^2}\right)$

- **Big Advantage:** Computing the exact posterior is computationally efficient!



Howard Raiffa
(1924 – 2016)



Robert Osher Schlaifer
(1914 – 1994)

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Normal Distribution: Representations

Scale-Location Parameters

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

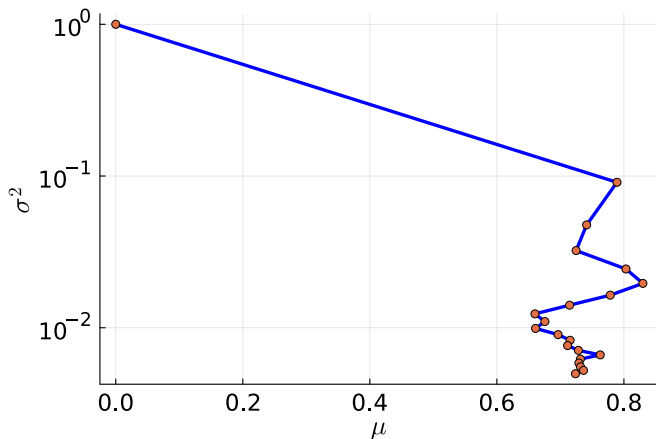
Conversions

$$\mathcal{N}(x; \mu, \sigma^2) = \mathcal{G}\left(x; \frac{\mu}{\sigma^2}, \frac{1}{\sigma^2}\right)$$

Two divisions only!

$$\mathcal{G}(x; \tau, \rho) = \mathcal{N}\left(x; \frac{\tau}{\rho}, \frac{1}{\rho}\right)$$

Posterior Inference

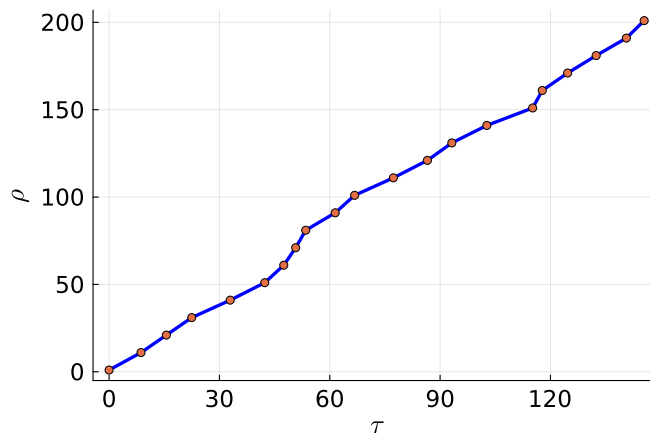


Natural Parameters

$$\mathcal{G}(x; \tau, \rho) = \sqrt{\frac{\rho}{2\pi}} \cdot \exp\left(-\frac{\tau^2}{2\rho}\right) \cdot \exp\left(\tau \cdot x - \rho \cdot \frac{x^2}{2}\right)$$

Conversions

Posterior Inference



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Normal Distributions: Efficient Products & Divisions

- **Theorem (Multiplication).** Given two one-dimensional Gaussian distributions $\mathcal{G}(x; \tau_1, \rho_1)$ and $\mathcal{G}(x; \tau_2, \rho_2)$ we have

$$\mathcal{G}(x; \tau_1, \rho_1) \cdot \mathcal{G}(x; \tau_2, \rho_2) = \mathcal{G}(x; \tau_1 + \tau_2, \rho_1 + \rho_2) \cdot \mathcal{N}(\mu_1; \mu_2, \sigma_1^2 + \sigma_2^2)$$

Gaussian density

Additive updates!

- **Theorem (Division).** Given two one-dimensional Gaussian distributions $\mathcal{G}(x; \tau_1, \rho_1)$ and $\mathcal{G}(x; \tau_2, \rho_2)$ where $\rho_1 > \rho_2$ we have

$$\frac{\mathcal{G}(x; \tau_1, \rho_1)}{\mathcal{G}(x; \tau_2, \rho_2)} = \frac{\mathcal{G}(x; \tau_1 - \tau_2, \rho_1 - \rho_2)}{\mathcal{N}(\mu_1; \mu_2, \sigma_2^2 - \sigma_1^2)} \cdot \frac{\sigma_2^2}{\sigma_2^2 - \sigma_1^2}$$

Correction factor

Subtractive updates!

Gaussian density

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Limit Normal Distributions: Dirac Delta and Uniform

- **Dirac Delta.** The Dirac delta function $\delta(\cdot)$ is defined as the limit $\sigma^2 \rightarrow 0$

$$\delta(x) = \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x; 0, \sigma^2)$$

- **Gaussian Uniform.** The Gaussian uniform $\mathcal{U}(\cdot)$ is defined as the limit $\sigma^2 \rightarrow \infty$

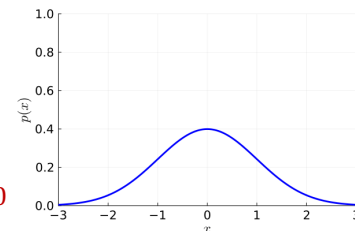
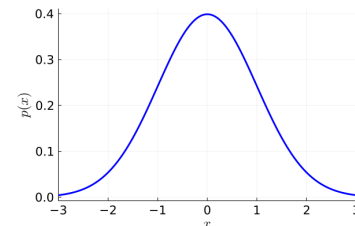
$$\mathcal{U}(x) = \lim_{\sigma^2 \rightarrow +\infty} \mathcal{N}(x; 0, \sigma^2)$$

- **Theorem (Convolution of Normal with Dirac).** For any $\mu \in \mathbb{R}$ and $\sigma^2 \in \mathbb{R}^+$

$$\int_{-\infty}^{+\infty} \delta(x) \cdot \mathcal{N}(x; \mu, \sigma^2) dx = \mathcal{N}(0; \mu, \sigma^2) \leftarrow \text{Gaussian density at } x = 0$$

- **Theorem (Product of Normal with Uniform).** For any $\mu \in \mathbb{R}$ and $\sigma^2 \in \mathbb{R}^+$

$$\frac{\mathcal{U}(x) \cdot \mathcal{N}(x; \mu, \sigma^2)}{\int_{-\infty}^{+\infty} \mathcal{U}(\tilde{x}) \cdot \mathcal{N}(\tilde{x}; \mu, \sigma^2) d\tilde{x}} = \mathcal{N}(x; \mu, \sigma^2) \leftarrow \text{Equivalent to multiplying with 1}$$



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Probability Distributions: Exponential Family

- **Exponential Family.** A family of distributions is said to belong to the exponential family if the probability density/mass function in terms of the parameterisation θ is

$$p(x) = \exp \left(\sum_i \theta_i \cdot T_i(x) - A(\theta) \right)$$

- The θ_i 's are called *natural parameters* and the T_i 's are called *sufficient statistics*.

Distribution $p(x)$	Natural Parameters θ	Sufficient Statistic $T(x)$
$\text{Bin}(x; n, \pi)$	$\log \left(\frac{\pi}{1 - \pi} \right)$	x
$\text{Beta}(\pi; \alpha, \beta)$	$[\alpha, \beta]$	$[\log(\pi), \log(1 - \pi)]$
$\mathcal{N}(x; \mu, \sigma^2)$	$\left[\frac{\mu}{\sigma^2}, \frac{1}{\sigma^2} \right]$	$\left[x, -\frac{x^2}{2} \right]$



Edwin Pitman
(1897 - 1993)



Georges Darmois
(1888 - 1960)



Bernhard Koopman
(1900 - 1991)

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- **Big Advantage:** Closed and efficient under multiplication (Bayes' rule!)

$$p(x; \theta_1) \cdot p(x; \theta_2) \propto p(x; \theta_1 + \theta_2)$$

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Maximum Likelihood/Maximum A-Posteriori Inference

1. **Maximum Likelihood.** Find the most *likely* function $f_{\text{ML}}(D)$ given the data D and approximate $p(f|D)$ by a single point distribution around

$$f_{\text{ML}}(D) = \underset{f}{\operatorname{argmax}} p(D|f)$$

2. **Maximum A Posterior.** Find the most *probable* function $f_{\text{MAP}}(D)$ given the data D and prior $p(f)$ and approximate $p(f|D)$ by a single point distribution around

$$f_{\text{MAP}}(D) = \underset{f}{\operatorname{argmax}} p(D|f) \cdot p(f)$$

■ **Pros:**

1. Learning = optimization in the hypothesis space (“gradient descent”)
2. Storing the model = storing the function parameters

■ **Cons:**

1. The posterior/likelihood is “peaked” around a single best predictor (convergence)
2. No model uncertainty after learning from data



Sir Ronald Fisher
(1890 – 1962)

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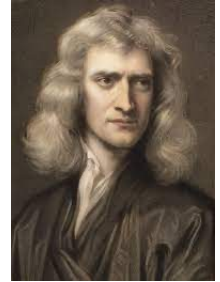
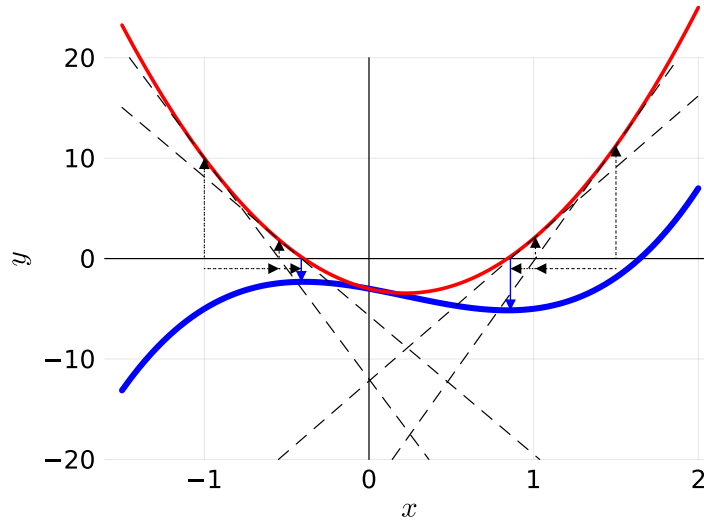
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Newton-Raphson Algorithm

- **Problem:** Find the local extrema of a function $f: \mathbb{R} \rightarrow \mathbb{R}$
- **Idea:** Find the zeros of the first derivative f' of the function!
- **Newton-Raphson Algorithm:**
Approximate f' at a point x_t with a linear function $g(x) = ax + b$ and find update x_{t+1} such that $g(x_{t+1}) = 0$

$$a = f''(x_t)$$
$$b = f'(x_t) - f''(x_t) \cdot x_t$$

$$x_{t+1} = -\frac{b}{a} = \frac{f''(x_t) \cdot x_t - f'(x_t)}{f''(x_t)}$$



Sir Isaac Newton
(1643 – 1727)

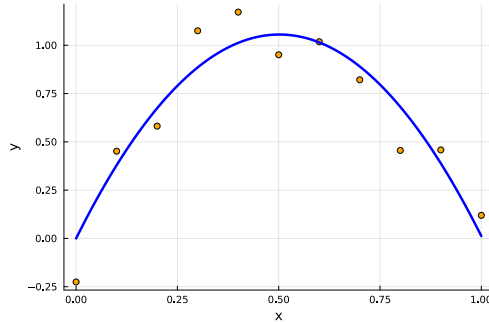
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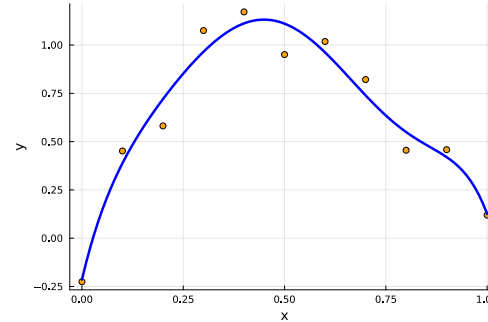
$$x_{t+1} = x_t - \frac{f'(x_t)}{f''(x_t)}$$

Maximum A-Posteriori Inference: Polynomial Regression

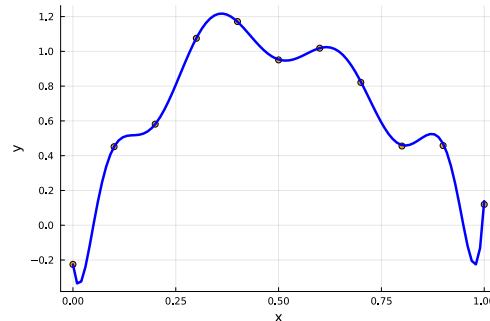
$$f(x) = w_1x + w_2x^2$$



$$f(x) = w_0 + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6$$



$$f(x) = \sum_{i=0}^{10} w_i \cdot x^i$$

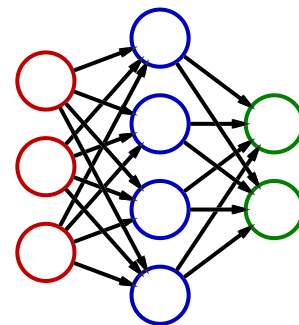


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Relation to Deep Learning

- **Deep Learning** is maximum likelihood inference on a layered function model
 - **Neural Networks:** $f(x) = h(W_L \cdots h(W_2 h(W_1 x)))$ where h is a sigmoid
 - Number of layers: L
 - Each element of each vector is called a “neuron”
 - Each product of the inner products is called a “synapse”
- **Maximum Likelihood** optimization via gradient descent (w.r.t. $W_1, W_2, \dots, W_L$)
 - Application of the chain rule of differentiation = back propagation
 - Predicting and gradient computations are matrix multiplications; today, they are sped up using GPUs (which parallelize matrix multiplication)
- **Regularization** for the Deep Learning algorithms are equivalent to prior assumptions on $p(W_1, W_2, \dots, W_L)$!



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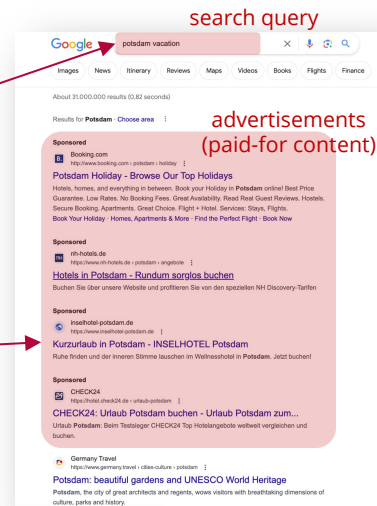
Key Question II: Decision Making and Prediction

- **Decision Making.** Given a training set $D \in (\mathcal{X} \times \mathcal{Y})^n$, a new input point $x \in \mathcal{X}$ and an action space $\mathcal{A}$, what action $\hat{a} \in \mathcal{A}$ shall be made at x based on D ?

- **Example:** Deciding which of n advertisement to show on a search result page
 - x_i contains all information of all advertisements i , the search query, the user, ...
 - $y_i \in \{0,1\}$ indicates, whether or not the advertisement is clicked by the user
 - D is the dataset of all displayed advertisements and whether or not they got clicked
 - $a \in \{1, \dots, n\}$ indicates, which one of the advertisements got chosen
- **Decision theory** is concerned with the theory of making decisions based on uncertain outcomes and assigning numerical consequences to the outcome

- **Prediction.** Given a training set $D \in (\mathcal{X} \times \mathcal{Y})^n$ and a new input point $x \in \mathcal{X}$, what prediction $\hat{y} \in \mathcal{Y}$ shall be made for the example x based on D ?

- **Observation 1:** Special case of decision making when $\mathcal{A} = \mathcal{Y}$
- **Observation 2:** In contrast to the predictive distribution $p(y|x, D)$ over **all** possible outcomes $y \in \mathcal{Y}$, we are committing to a **specific** outcome $\hat{y}(x, D) \in \mathcal{Y}$ in **prediction**!



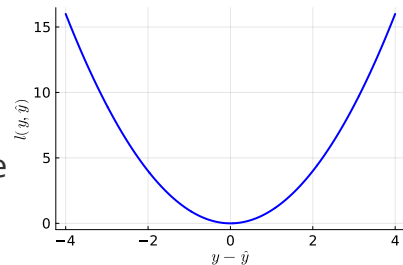
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Loss Functions

- **Problem:** What is the consequence of each action $\hat{a} \in \mathcal{A}$ or $\hat{y} \in \mathcal{Y}$ given that the truth was $y \in \mathcal{Y}$?
- **Loss Function.** A loss function $l: \mathcal{Y} \times \mathcal{A} \rightarrow \mathbb{R}$ is a function mapping the outcome space $\mathcal{Y}$ and an action space $\mathcal{A}$ to a real number representing the "cost" associated with taking the action $a \in \mathcal{A}$ when the true state of the world is $y \in \mathcal{Y}$.
 - Losses are given by the domain problem; there are no "true" losses!
 - **Decision Making Example 1:** Deciding which of n advertisement to show
 - When advertisement $a \in \{1, \dots, n\}$ is chosen and the click happens ($y_a = 1$), then the utility is the bid amount b_a being paid; all other advertisers do not pay
 - **Decision Making Example 2:** Giving a treatment after a cancer test
 - **Prediction Example:** Predicting the temperature in Potsdam
 - If we predict too high is as bad as too low and losses grow faster than the difference

		Actions	
		treat	nothing
Outcomes	Cancer	0	1000
	No cancer	1	0



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$$l(y, \hat{y}) = (y - \hat{y})^2$$

Optimal Decisions: Expected Loss Minimization Principle

- **Expected Loss Minimization.** Given a predictive model $p(y|x, D)$ and a loss function $l: \mathcal{Y} \times \mathcal{A} \rightarrow \mathbb{R}$, the optimal action $a_{\text{opt}}(x)$ is minimizing the expected loss

$$a_{\text{opt}}(x) := \operatorname{argmin}_{a \in \mathcal{A}} E_{y \sim p(y|x, D)}[l(y, a)]$$

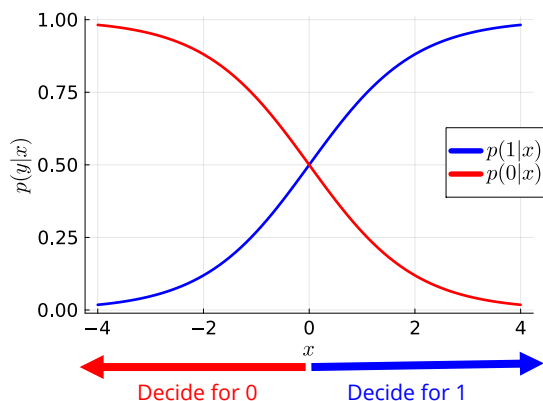
- Optimal decisions require (yet again) solving an optimization problem!
- **Decision Making Example:** Deciding which of n advertisement to show
 - Each expected loss is equal to $E_{y_a \sim p(y_a|x_a, D)}[-b_a \cdot y_a] = -P(y_a = 1|x, D) \cdot b_a$
 - The add with the largest expected bid $P(y_a = 1|x, D) \cdot b_a$ should be shown!
- **Prediction Example:** For squared loss $y_{\text{opt}} = E_{y \sim p(y|x, D)}[y]$
 - **Proof:** Taking the first derivative of $E_{y \sim p(y|x, D)}[l(y, \hat{y})]$ and setting it to zero gives

$$\begin{aligned} \frac{d}{d\hat{y}} E_{y \sim p(y|x, D)}[l(y, \hat{y})] &= \sum_y p(y|x, D) \cdot \frac{d}{d\hat{y}} (y - \hat{y})^2 \\ 0 &= \sum_y p(y|x, D) \cdot (2 \cdot (y - y_{\text{opt}})) \\ 0 &= 2 \cdot \left(\sum_y p(y|x, D) \cdot y - \sum_y p(y|x, D) \cdot y_{\text{opt}} \right) \\ 0 &= E_{y \sim p(y|x)}[y] - y_{\text{opt}} \end{aligned}$$

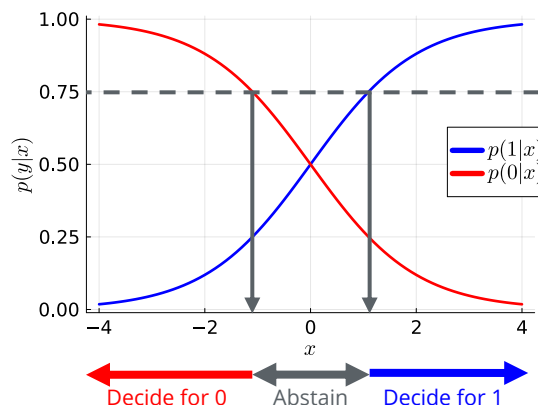
Decision Making under Uncertainty

- **Problem:** In practice, we rarely have predictive distributions $p(y|x, D)$ which are concentrated on a single outcome $\hat{y} \in \mathcal{Y}$.
- **Idea:** Introduce the action “to abstain” from making a prediction (and use human intelligence!)
- **Example:** Automatically deciding on a treatment (health) or job application (business)

**Optimal Binary Prediction
(without abstaining)**



**Optimal Binary Prediction
(with abstaining)**



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1. Inference Methods

- Inference is the task of inferring what we know about the plausibility of a prediction function in light of training data
- Bayesian Inference is the only consistent inference technique, but it requires huge summations which is (usually) computationally too hard
- Maximum Likelihood Estimation is often easier and reduces machine learning to parameter optimization – but we are losing model uncertainty

2. Decision Making

- Decision making solves the second big problem: making automated decisions based on future predictions
- We *always* require domain-specific loss functions
- Except in a few special cases (e.g., squared loss), automated decision making requires heavy optimization (again!)
- We will not dive deeper into decision making methods in the rest of the lecture

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See you next week!